STAGE 1 · V-PCC DELIVERY OVER RELIABLE QUIC

transport path under test — impairment injected here

Volumetric server

Caddy · HTTP/3 origin
V-PCC GOF chunks
TLS 1.3 · AES-128-GCM

QUIC

Impairment

Linux TC: TBF + netem
one UDP port per slice
loss · added latency

HTTP/3

Edge renderer

Unity 6 · V3C decoder
RTX 4070 · ALVR
per-GOF telemetry

ALVR

Impairment slices

one UDP port each on the same origin.
2×2 factorial on B, C, D at a fixed encoder;
A anchors the clean ceiling Qc.

A	**Clean**	no loss, no delay	**RA · GOF 32**
B	**Loss**	loss only	**LDP · GOF 4**
C	**Latency**	delay only	**LDP · GOF 4**
D	**Loss + latency**	both	**LDP · GOF 4**

STAGE 2 · EDGE-TO-HMD

private 5G, held constant

Private 5G

Ericsson SA
RU · BBU · NC
colocated edge

NR

Quest 3

5G USB dongle
ACR rating
no Wi-Fi

Stage 2 is transparent

Rendering happens at the edge; the headset receives finished stereo frames. The casting link is identical in every condition, so any MOS difference originates in Stage 1.